

Predictive Fault Detection Report

LSTM-based anomaly detection and failure classification

Generated 11 Apr 2026, 22:53

Dataset	predictive_maintenance.csv	Input Shape	(10, 6)	Classes	6
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Run Overview

Field	Value
Generated	2026-04-11 22:53:24
Dataset	predictive_maintenance.csv
Input Shape	(10, 6)
Classes	Heat Dissipation Failure, No Failure, Overstrain Failure, Power Failure, Random Failures, Tool Wear Failure

Run Settings

Setting	Value
Epochs	1
Batch Size	64
Time Steps	10
Simulator	Disabled
Simulation Steps	200

Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Binary LSTM	0.9804	0.0000	0.0000	0.0000	0.8285
Multiclass LSTM	0.9699	0.9708	0.9699	0.9700	N/A

Binary Accuracy	0.980	Multiclass Accuracy	0.970	Elapsed	0.3 min
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Binary LSTM — Visualizations

Binary LSTM — Training History

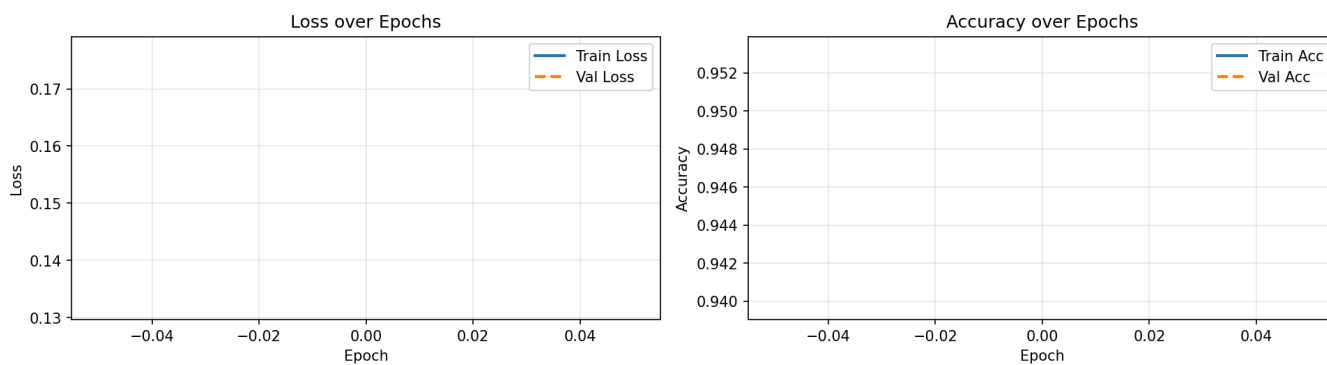


Figure 1 — Binary LSTM: Training Loss & Accuracy

Binary LSTM — Confusion Matrix

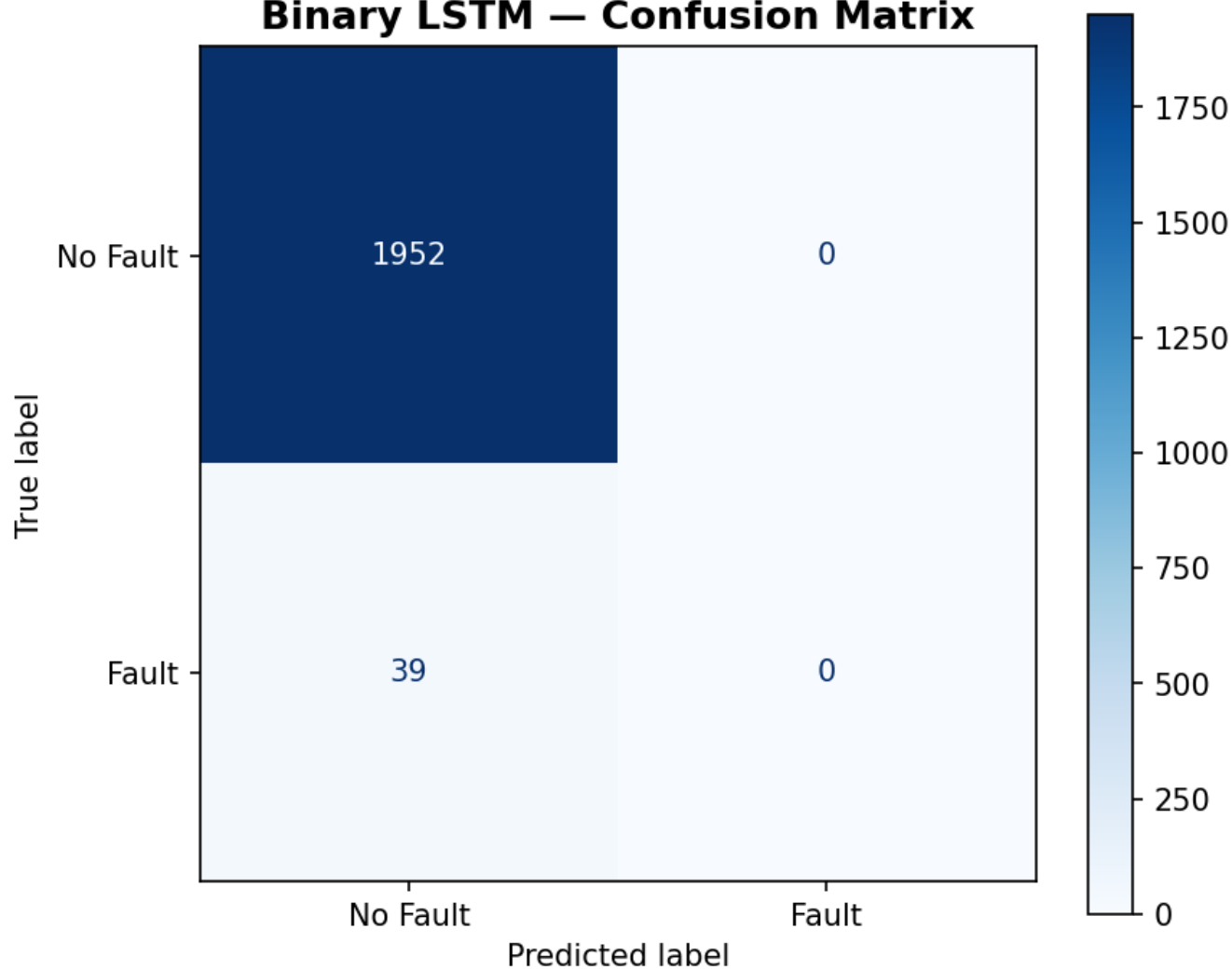


Figure 2 — Binary LSTM: Confusion Matrix

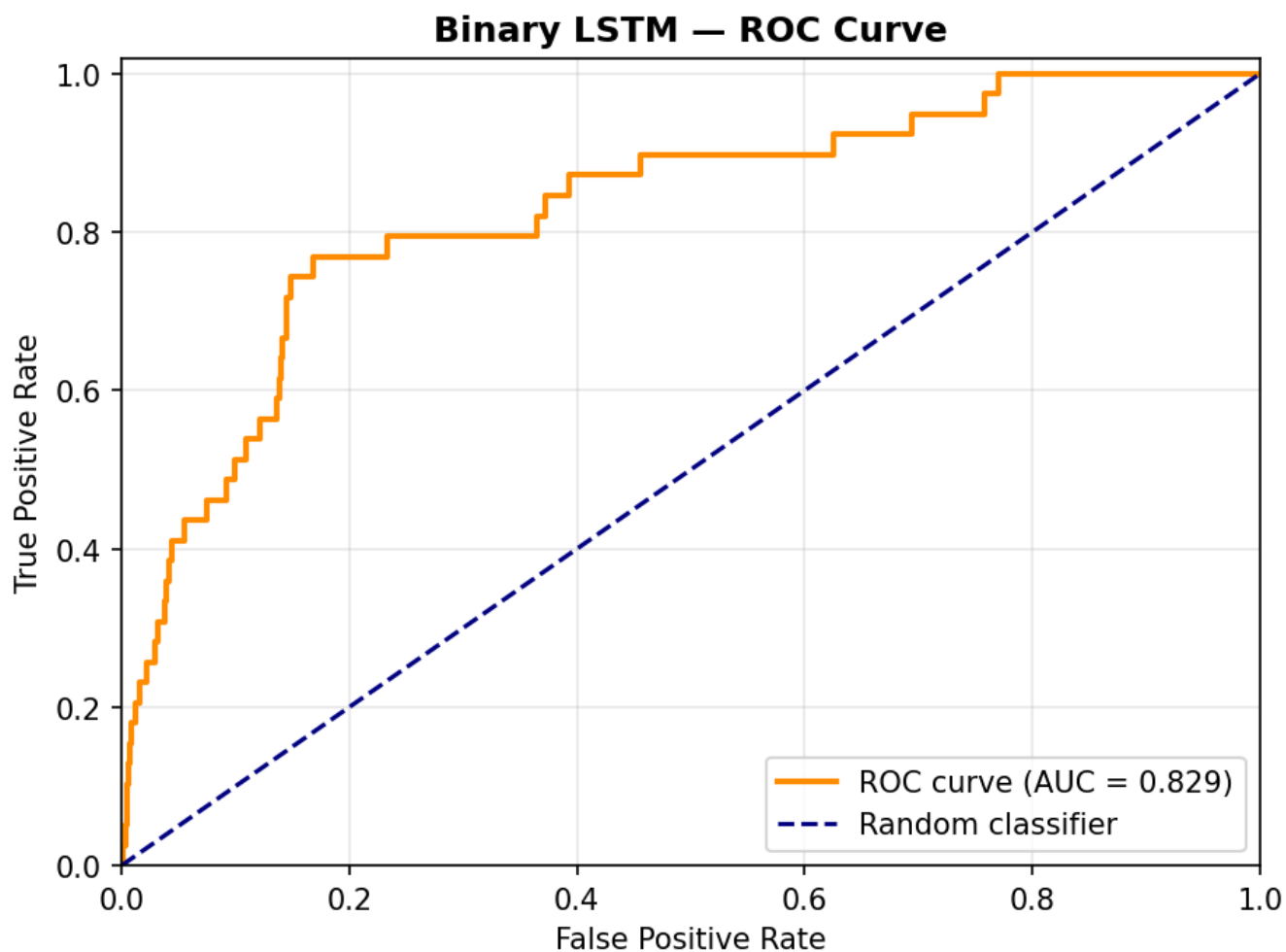


Figure 3 — Binary LSTM: ROC Curve

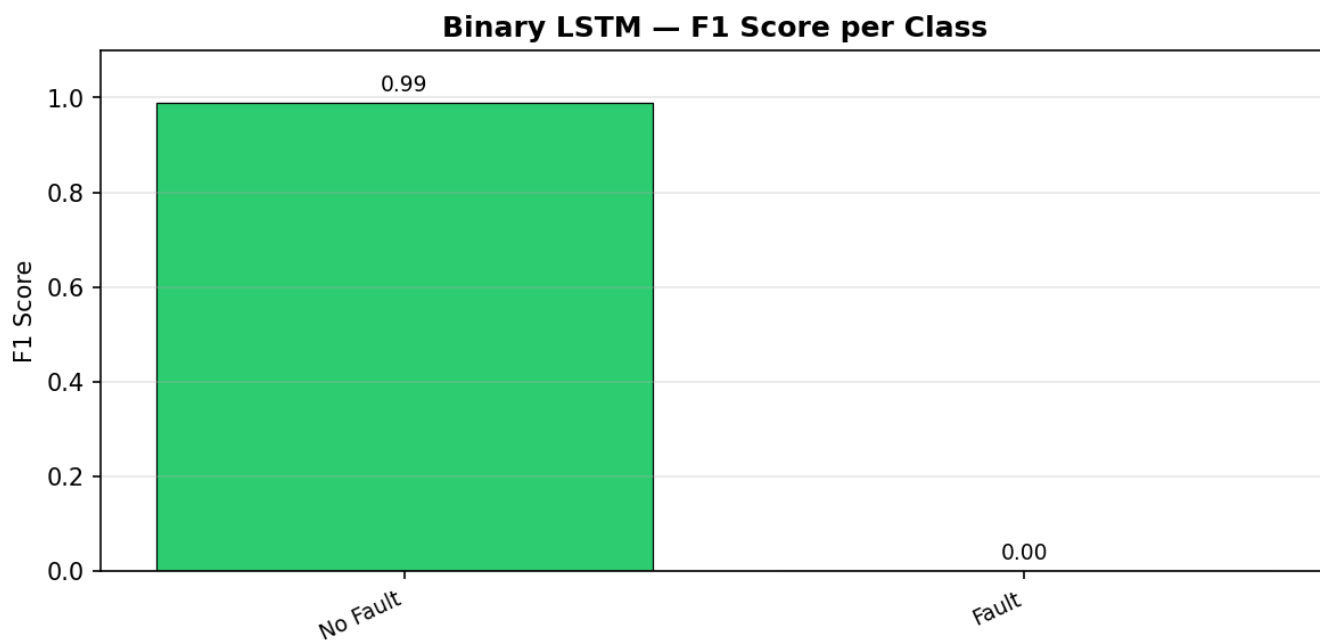


Figure 4 — Binary LSTM: F1 Score per Class

Multiclass LSTM — Visualizations

Multiclass LSTM — Training History

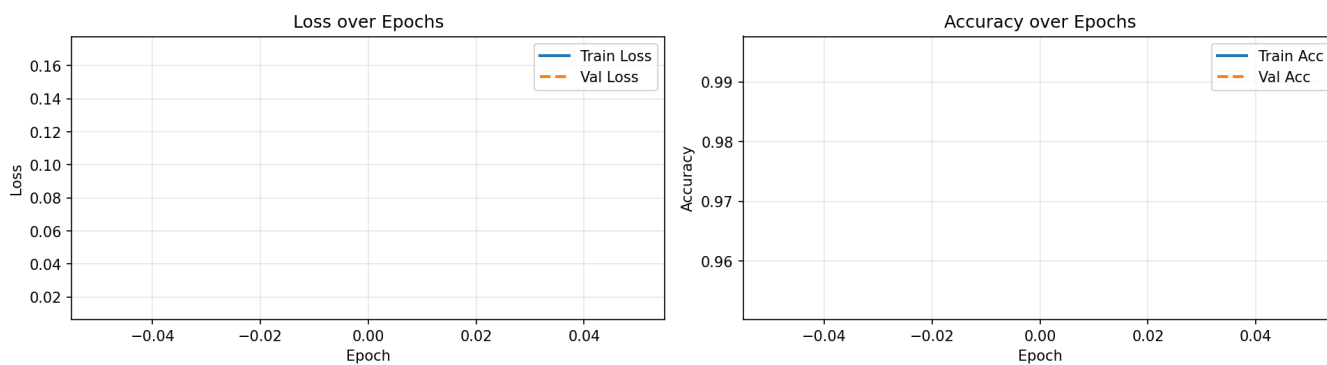


Figure 5 — Multiclass LSTM: Training Loss & Accuracy

Multiclass LSTM — Confusion Matrix

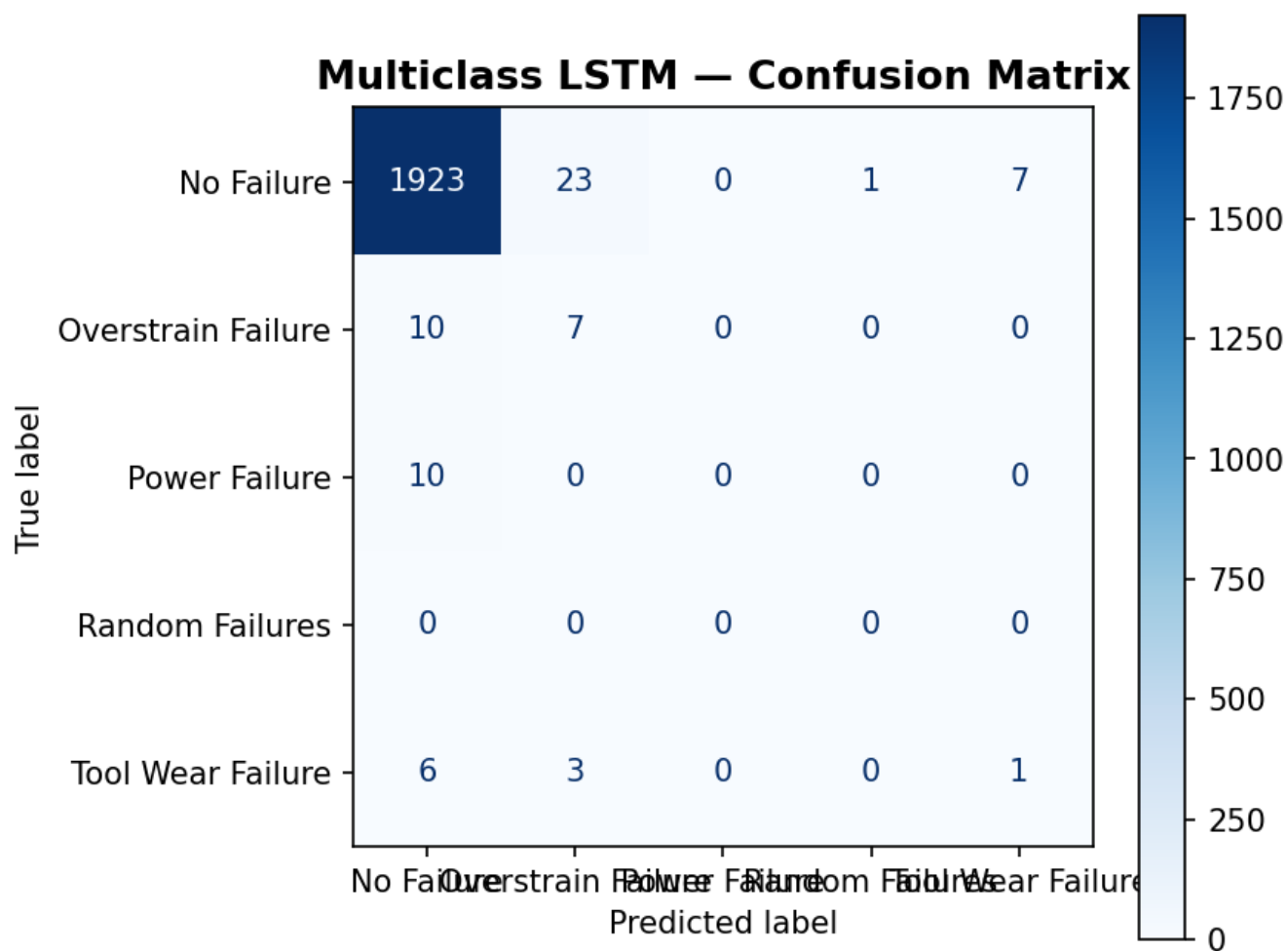


Figure 6 — Multiclass LSTM: Confusion Matrix

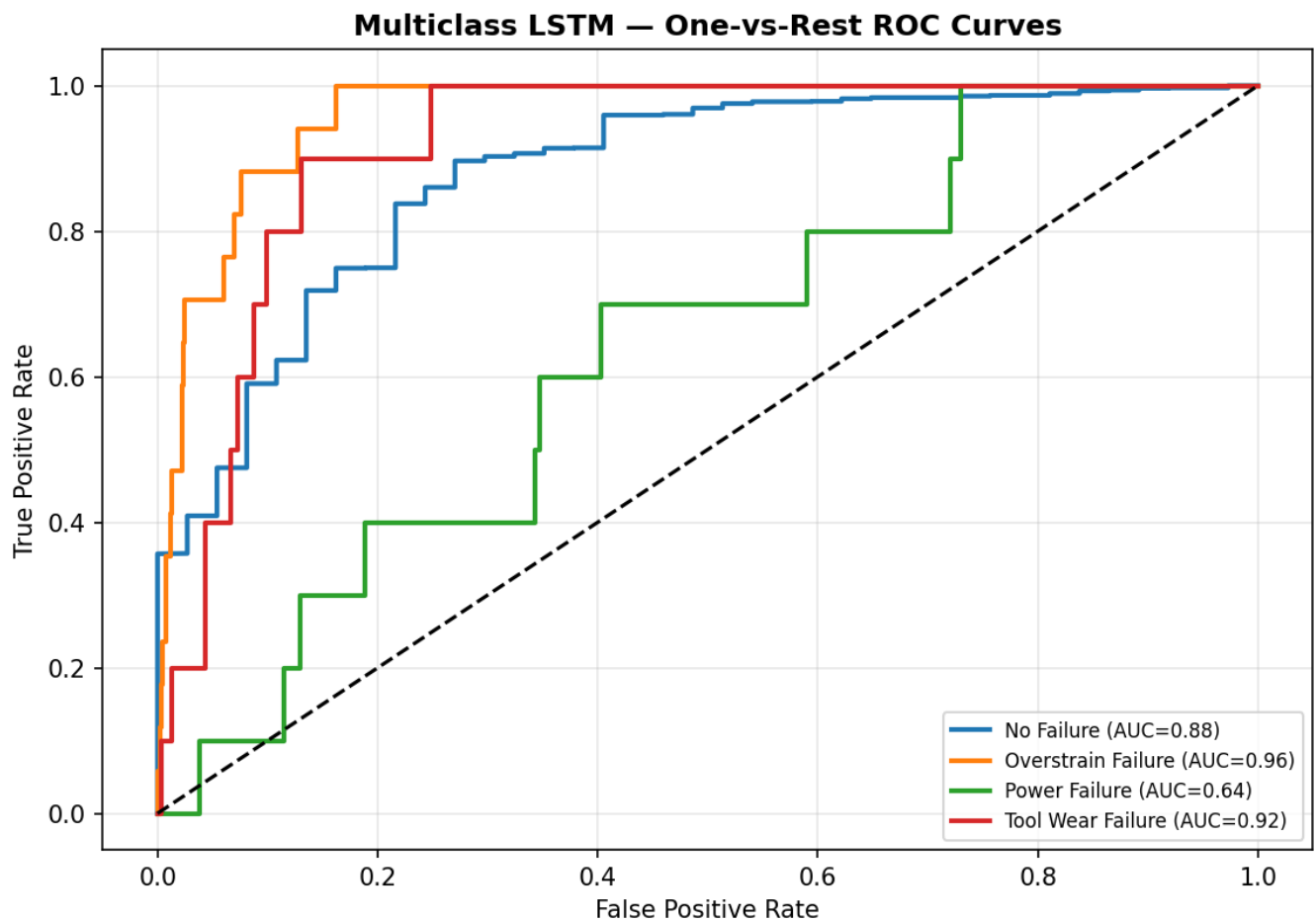


Figure 7 — Multiclass LSTM: One-vs-Rest ROC Curves

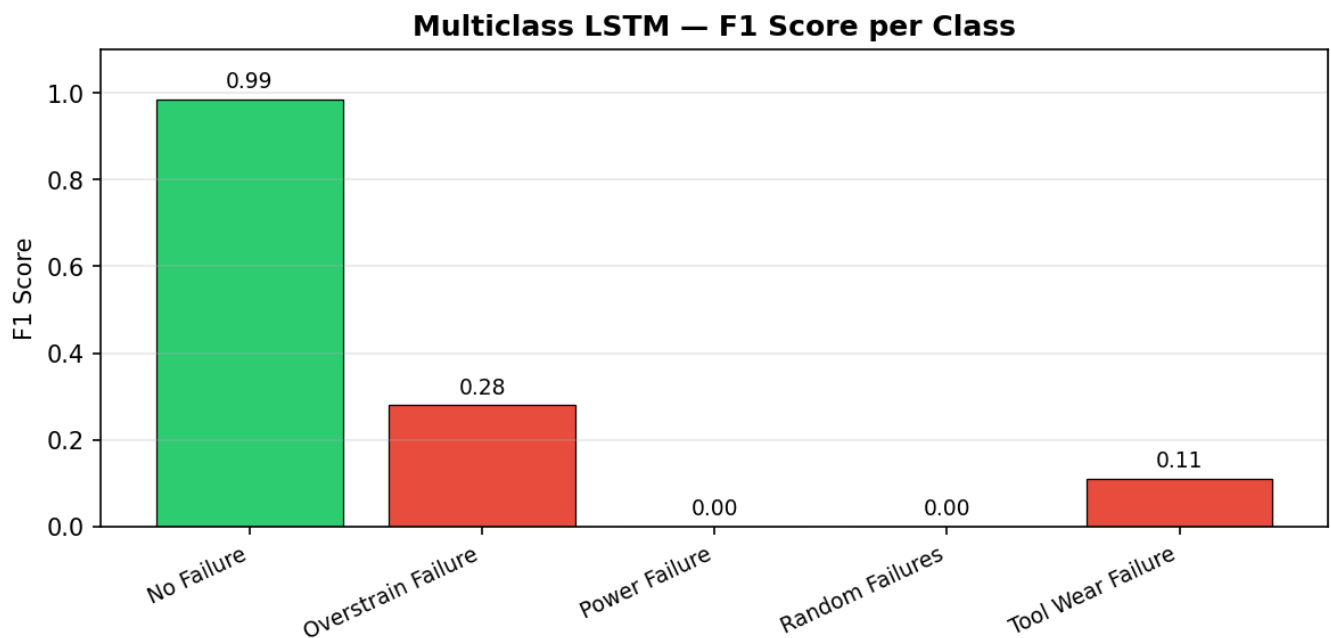


Figure 8 — Multiclass LSTM: F1 Score per Class

Generated Files

Status	File	Description
OK	01_Binary_LSTM_Training_History.png	Binary LSTM training curves

OK	02_Binary_LSTM_Confusion_Matrix.png	Binary confusion matrix
OK	03_Binary_LSTM_ROC_Curve.png	Binary ROC curve
OK	04_Binary_LSTM_F1_Scores.png	Binary F1 scores per class
OK	05_Multiclass_LSTM_Training_History.png	Multiclass LSTM training curves
OK	06_Multiclass_LSTM_Confusion_Matrix.png	Multiclass confusion matrix
OK	07_Multiclass_LSTM_ROC_Curves.png	Multiclass ROC curves (one-vs-rest)
OK	08_Multiclass_LSTM_F1_Scores.png	Multiclass F1 scores per class
OK	results_summary.csv	Combined metrics summary
SKIPPED	simulation_log.csv	Robot simulator step-by-step log
OK	Report.pdf	Full PDF report with embedded plots

Total elapsed time: 0.3 min